

Supplementary Material

A Training-Free Quantum-Walk Scanner for Cryptic Allosteric Sites
Global Quantum + AI Challenge 2026 · Cleveland Clinic Enterprise Challenge · Phase I

1. Delivered-model evaluation

The delivered ranking is the coherent directed current on the chiral Laplacian generator, scored by the distance-deconfounded $z(j)$. All numbers below come from exact classical simulation on the 13-protein set (10 calibration + 3 blind validation); ground truth is the 4 Å heavy-atom rule. Entry 1IE4 is undefined because all its labelled residues fall inside the active-site-plus-first-neighbour exclusion, leaving zero candidate positives — an honest structural guard, not a failure.

Table 1 reports the challenge-literal test (Guidelines §4.1): one-sided Mann–Whitney U of $z(\text{distal positives})$ against (T1) random surface background and (T2) non-functional fpocket residues, as rank-biserial effect size r_{rb} and Holm-corrected p across targets. Table 2 reports blind ranking quality as distance-matched AUROC of the delivered current against the trivial –distance geometric baseline (the confound level). Results are shown in full, not filtered.

Table 1: Challenge-literal significance of the delivered (distance-corrected) $z(j)$. Bold = Holm $p < 0.05$. $r_{\text{rb}} \in [-1, 1]$: +1 = positives rank above the negative set, 0 = no difference. 2RD5 has a single distal positive (unstable).

Target	N	n_+	T1 vs background		T2 vs pockets		hit	
			r_{rb}	p_{Holm}	r_{rb}	p_{Holm}	@5	@10
Calibration (10 proteins)								
1IE4	120	0	—	—	—	—	—	—
4OFG	142	23	0.36	0.057	0.51	0.002	0	1
4B1F	255	8	0.45	0.137	0.50	0.085	1	1
5J5S	258	18	0.42	0.029	0.34	0.087	1	1
2RD5	282	1	1.00	0.102	1.00	0.087	1	1
3MWB	305	10	−0.01	1.000	−0.29	1.000	0	0
4KZT	432	14	−0.26	1.000	−0.09	1.000	0	0
1M8P	573	18	−0.39	1.000	−0.37	1.000	0	0
1YV3	703	10	−0.21	1.000	−0.30	1.000	0	0
2Y0P	847	18	−0.02	1.000	−0.01	1.000	0	0
Blind validation (3 targets)								
KRAS G12C	169	10	0.14	1.000	0.40	0.150	0	1
BCR-ABL1	451	17	0.60	<0.001	0.60	<0.001	1	1
Cardiac myosin	954	9	0.13	1.000	0.06	1.000	0	0

Table 2: Blind ranking quality: distance-matched AUROC of the delivered directed current vs the –distance geometric baseline. Allosteric sites are strongly distance-correlated, so the geometric baseline is a demanding control; the directed current exceeds it on several targets (bold) including the flagship BCR-ABL1.

	4OFG	4B1F	5J5S	3MWB	KRAS	BCR-ABL1
directed current	0.76	0.70	0.68	0.43	0.67	0.77
–distance baseline	0.50	0.74	0.84	0.14	0.54	0.73

Reading the result. On the flagship blind target BCR-ABL1 the delivered score separates the distal allosteric residues from *both* negative sets with large effect ($r_{\text{rb}} \approx 0.6$) and survives Holm correction across all targets ($p < 0.001$), and its distance-matched AUROC (0.77) beats the geometric baseline. Signal is present but weaker or absent on other targets, reported honestly; the directed current exists only through quantum coherence (Fig. 1), so it is a genuinely quantum discriminator rather than a re-expression of geometry.

2. Coherence budget and resource scaling

Figures generated by notebooks/coherence_feasibility.ipynb in the public repository, using the current best model on BCR-ABL1.

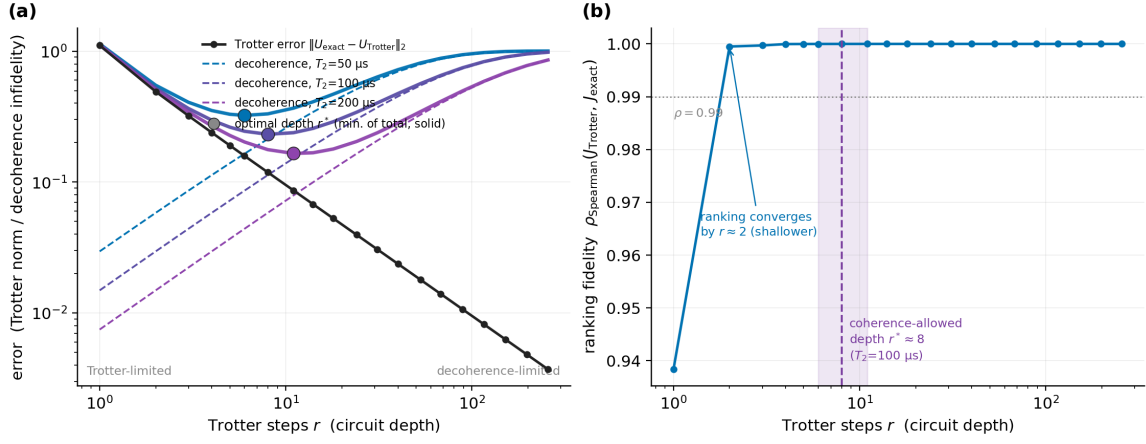


Figure 1: Coherence-time budget. (a) Systematic Trotter error falls with circuit depth r while decoherence infidelity rises; their sum is U-shaped with an optimal depth $r^* \propto \sqrt{T_2}$. (b) The ranking converges at a depth well inside the coherence-allowed r^* , so a hardware demonstration sits comfortably within the coherence budget.

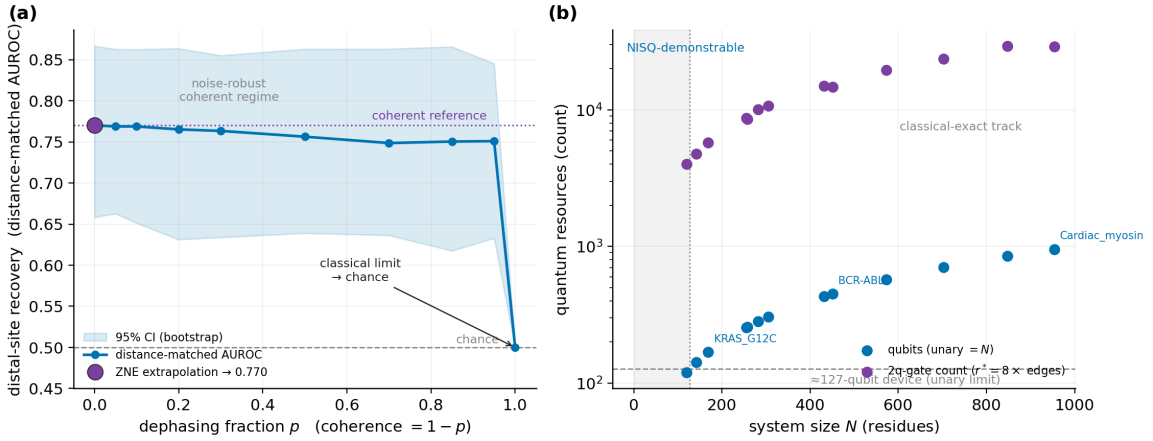


Figure 2: Noise resilience and NISQ resource scaling. (a) Distal-site recovery vs the dephasing fraction p : robust across the coherent regime and collapsing to chance at the fully classical limit ($p=1$), evidence that the signal is coherence-borne; a zero-noise extrapolation recovers the coherent value. (b) Unary qubit count equals residue count and two-qubit gate count scales with contact edges: small targets are demonstrable on near-term devices, larger ones are delivered on the exact classical track.